INSPIRE AI

HOMEWORK

Build and Test a CRUD API



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Templates keep inputs for future quotes

> As an operator, I want to create, read, update and delete templates of common parcel inputs so that I can maintain inputs for future quotes.

Original request · US-07

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Continue in your Day 6 homework repository

Your Day 6 classroom project in your personal homework repository

Before class, your personal Day 6 homework repository is initialized from the approved parcel-quote main snapshot. Clone your assigned repository and use it throughout the day. For homework, continue from your own classroom quote API work in that same repository. No template scaffold is supplied. Use the quote code and tests as references to build the template resource yourself, preserving your classroom code and commit history. Run ./mvnw test on macOS or Linux, or .\mvnw.cmd test in Windows PowerShell, before editing. Record any unfinished classroom behavior or failing baseline before starting the template task.

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Homework: quote template CRUD

Independently describe, implement and test create, read, replacement and delete for one in-memory resource in the existing case.

1. Check Your Classroom Baseline and the Template Examples
2. Write the API Contract
3. Test and Implement Create and Read
4. Test and Implement Update and Delete
5. Run Regression and Submit

Produce

OpenAPI contract

Runnable project and API tests

TDD and regression results

60 minutes · whole activity

Source activity: d06-hw

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Template data and the resource boundary

US-07-AC01	A quote template contains id, name, weight_kg, zone, and service. Store templates in a separate in-memory repository for the current process. Responses contain these five fields and no amount_cents. Template management does not calculate prices, check service availability, create quotes, or apply a template to a quote.

Selected criteria for US-07 · rules, not recorded test results

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Shared field value rules

US-01-AC01	Weight must be an integer from 1 to 20 kg. Zone must be local or remote. Service must be standard or express. Reject invalid inputs without a quote.

Selected criteria for US-01 · rules, not recorded test results

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Creation and reading by ID

US-07-AC02	POST /quote-templates accepts name, weight_kg, zone, and service. Store a valid template under a server-generated id and return status 201 with the stored template. Each successful create gets a new id; template names need not be unique.
US-07-AC03	GET /quote-templates/{id} returns status 200 and all five stored fields for an existing id. An unknown id returns status 404. Reading does not change templates or quotes.

Selected criteria for US-07 · rules, not recorded test results

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Replacement and deletion

US-07-AC04	PUT /quote-templates/{id} fully replaces the four input fields of an existing template and returns status 200 with the updated template and the same id. An unknown id returns status 404 and creates no template. Rejection leaves all stored fields unchanged.
US-07-AC05	DELETE /quote-templates/{id} removes the existing template and returns status 204 with an empty response body. A later GET for that id returns status 404. Deleting an unknown id returns status 404. Other templates remain unchanged.

Selected criteria for US-07 · rules, not recorded test results

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Input rejection and its stored-state rules

US-07-AC06	Missing or wrongly typed input fields and client-supplied id or amount_cents return status 400. A name containing no non-whitespace characters, or a value outside the shared input rules in US-01-AC01, returns status 422 with code INVALID_INPUT. Invalid creates add no template; invalid updates leave the template unchanged. For PUT, check whether the id exists before validating the input.

Selected criteria for US-07 · rules, not recorded test results

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Other templates and quotes stay unchanged

US-07-AC07	Template operations leave other templates and every existing quote unchanged. Quote calculation, creation, update, read, and their current contracts continue to work. Template persistence, a list operation, a UI, and template-to-quote links are outside this change.

Selected criteria for US-07 · rules, not recorded test results

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Creating and reading the same template

POST /quote-templates, then GET /quote-templates/{id} using the returned id.

Given

Templates	Empty

Request body

Name	Weight (kg)	Zone	Service
Local standard	3	local	standard

US-07-E01 · Specification expectation · Actual result: not recorded

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Creating and reading the same template

POST /quote-templates, then GET /quote-templates/{id} using the returned id.

Then · Expected result

POST status	201
ID	generated; use the returned id rather than a fixed number
GET status	200

Stored fields

Name	Weight (kg)	Zone	Service
Local standard	3	local	standard

Response fields	id, name, weight_kg, zone, service
Template count	1

US-07-E01 · Specification expectation · Actual result: not recorded

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The same name can produce two templates

POST the same request to /quote-templates twice.

Given

Templates	Empty

Request body

Name	Weight (kg)	Zone	Service
Local standard	3	local	standard

Then · Expected result

Status for each request	201
IDs	two different server-generated values
Template count	2

US-07-E09 · Specification expectation · Actual result: not recorded

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Replacing all four input fields

PUT /quote-templates/17, then GET /quote-templates/17.

Given

Template

ID	Name	Weight (kg)	Zone	Service
17	Local standard	3	local	standard

Request body

Name	Weight (kg)	Zone	Service
Remote express	10	remote	express

US-07-E02 · Specification expectation · Actual result: not recorded

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Replacing all four input fields

PUT /quote-templates/17, then GET /quote-templates/17.

Then · Expected result

PUT status	200
GET status	200

Template

ID	Name	Weight (kg)	Zone	Service
17	Remote express	10	remote	express

Template count	1

US-07-E02 · Specification expectation · Actual result: not recorded

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A blank name preserves the old template

PUT /quote-templates/17, then GET /quote-templates/17.

Given

Template

ID	Name	Weight (kg)	Zone	Service
17	Local standard	3	local	standard

Request body

Name	Weight (kg)	Zone	Service
" "	5	remote	express

US-07-E03 · Specification expectation · Actual result: not recorded

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A blank name preserves the old template

PUT /quote-templates/17, then GET /quote-templates/17.

Then · Expected result

PUT status	422
Error code	INVALID_INPUT
GET status	200

Template

ID	Name	Weight (kg)	Zone	Service
17	Local standard	3	local	standard

Template count	1

US-07-E03 · Specification expectation · Actual result: not recorded

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Deletion preserves other resources

DELETE /quote-templates/17, then GET each template by its id and GET /quotes/17.

Given

Existing quote

ID	Weight (kg)	Zone	Service	Amount (cents)
17	3	local	standard	1300

Templates

ID	Name	Weight (kg)	Zone	Service
17	Local standard	3	local	standard
23	Remote standard	5	remote	standard

US-07-E04 · Specification expectation · Actual result: not recorded

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Deletion preserves other resources

DELETE /quote-templates/17, then GET each template by its id and GET /quotes/17.

Then · Expected result

DELETE status	204
DELETE body	Empty body
GET of deleted ID	404
GET of other ID	200

Remaining template

ID	Name	Weight (kg)	Zone	Service
23	Remote standard	5	remote	standard

Template count	1

US-07-E04 · Specification expectation · Actual result: not recorded

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The original quote remains unchanged

DELETE /quote-templates/17, then GET each template by its id and GET /quotes/17.

Then · Expected result

GET of quote 200

Existing quote

ID	Weight (kg)	Zone	Service	Amount (cents)
17	3	local	standard	1300

US-07-E04 · Specification expectation · Actual result: not recorded

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Unknown IDs leave the template unchanged

From a fresh copy of the given state for each request, GET, PUT, and DELETE /quote-templates/999.

Given

Template

ID	Name	Weight (kg)	Zone	Service
17	Local standard	3	local	standard

Unknown ID

999

PUT body

Name	Weight (kg)	Zone	Service
Remote standard	5	remote	standard

US-07-E05 · Specification expectation · Actual result: not recorded

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Unknown IDs leave the template unchanged

From a fresh copy of the given state for each request, GET, PUT, and DELETE /quote-templates/999.

Then · Expected result

Status for each request	404

Template

ID	Name	Weight (kg)	Zone	Service
17	Local standard	3	local	standard

Template count	1

US-07-E05 · Specification expectation · Actual result: not recorded

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An invalid zone adds no template

POST /quote-templates.

Given

Templates	Empty

Request body

Name	Weight (kg)	Zone	Service
Invalid zone	2	unknown	standard

Then · Expected result

Status	422
Error code	INVALID_INPUT
Template count	0

US-07-E06 · Specification expectation · Actual result: not recorded

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A missing name returns 400

POST /quote-templates without name.

Given

Templates	Empty

Request body

Weight (kg)	Zone	Service
3	local	standard

Then · Expected result

Status	400
Template count	0

US-07-E07 · Specification expectation · Actual result: not recorded

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Templates store inputs without a price

POST /quote-templates, then read the original quote through GET /quotes/17.

Given

Existing quote

ID	Weight (kg)	Zone	Service	Amount (cents)
17	3	local	standard	1300

Templates

Empty

Request body

Name	Weight (kg)	Zone	Service
Remote express inputs	11	remote	express

US-07-E08 · Specification expectation · Actual result: not recorded

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Templates store inputs without a price

POST /quote-templates, then read the original quote through GET /quotes/17.

Then · Expected result

POST status	201
Template ID	generated
Stored fields	Name: Remote express inputs; Weight (kg): 11; Zone: remote; Service: express
Response fields	id, name, weight_kg, zone, service

US-07-E08 · Specification expectation · Actual result: not recorded

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The original quote remains unchanged

POST /quote-templates, then read the original quote through GET /quotes/17.

Then · Expected result

GET of quote	200				
Existing quote	ID	Weight (kg)	Zone	Service	Amount (cents)
	17	3	local	standard	1300
Quote count	1				

US-07-E08 · Specification expectation · Actual result: not recorded

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Check your code and write the contract

Check Your Classroom Baseline and the Template Examples

Stay in your personal Day 6 homework repository. Read the template requirements and examples, locate useful patterns in your quote code and tests, and run the classroom regression baseline. Record setup blockers and unfinished classroom behavior separately from the new template work.

Write the API Contract

Add template schemas and the four operations, request fields, success and error responses, and examples to your existing openapi.yaml. Preserve the quote contract and validate the document with ./mvnw -Dtest=OpenApiTest test or .\mvnw.cmd -Dtest=OpenApiTest test on Windows.

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Test template creation and reading

Test and Implement Create and Read

Use your classroom MockMvc tests as references. Write API assertions from the creation and read examples, inspect their initial result, then implement the template types, separate in-memory storage, generated ids, validation and smallest passing behavior. Keep tests independent and check invalid requests and separate creates using the returned ids.

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Replacement, deletion and regression

Test and Implement Update and Delete

Add direct template seed and reset setup for independent update and delete tests. Implement each behavior after checking its initial test result. Verify unchanged ids, rejected updates, unknown ids, the empty deletion response and preservation of another template.

Run Regression and Submit

Validate OpenAPI and run template API and existing quote regression checks. Record commands, actual initial and later results for at least one new behavior, final results and checked version in records/api-tests.md. Commit and push the project, tests and evidence in the same personal homework repository, preserving classroom commits, and label unfinished work.

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Submit the OpenAPI contract

OpenAPI contract

`openapi.yaml`

- Define the template schemas and selected requests, responses and concrete examples. Preserve the existing quote contract.

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Submit runnable code and API tests

Runnable project and API tests

- Include production changes and tests, retain the Maven wrapper and supplied quote behavior, and identify selected examples in test names or comments.
- Equivalent examples may share a test. Assert returned fields and relevant stored state as well as status codes; test order must not supply state.

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Submit the actual test evidence

TDD and regression results

`records/api-tests.md`

- Include commands and actual initial and later results for at least one new behavior, final template and quote regression results, and the checked code version. Keep this a short run record.
- Identify any setup blocker, supplied evidence, or unfinished behavior; distinguish local in-memory API evidence from database storage.

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Completion: contract and behavior

- The validated OpenAPI file describes the four required operations, request fields, success and rejection responses, and concrete examples consistent with the selected behavior.
- The runnable project contains the learner's four template operations and API assertions covering the assigned examples, including unchanged data after rejection, deletion isolation, and unchanged quote behavior. Template types, separate storage, generated ids, validation and test setup are implemented by the learner; classroom code and commits remain preserved in the same personal Day 6 homework repository.

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Completion: traceable test results

- Identify examples in test names or comments. Include the actual initial and later result for at least one new behavior, final template and quote regression results, commands and checked code version. Label blocked or unfinished work; a passing check is not evidence of database storage.

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Build the template resource yourself

- Complete this coding task individually without AI assistance.
- Build the template types, separate in-memory storage, generated ids, validation, test reset/seed setup, schemas and four operations yourself. Use your classroom quote code and tests as references; no template scaffold is supplied.
- Keep templates in memory and preserve existing quotes. Database persistence, price calculation, service availability checks, listing, UI work and applying a template to a quote are outside this task.

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The time limit and remaining constraints

- Use the taught MockMvc setup with real collaborators. Do not introduce Mockito, Playwright or CI configuration.
- Use the same personal Day 6 homework repository for classroom work and this task; preserve classroom commits. Later prepared classroom copies stay separate from this submission. Stop at 60 minutes and label unfinished work.

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The course AI policy for reference

- Learners do most engineering work without AI. Sessions marked controlled-required may use it within their stated scope. Day 1 homework also permits the provided business requester Agent for closed questions only. Learners write their own examples and retain the dialogue; other AI assistance is not allowed for that homework. Day 10 permits bounded assistance for selected test moves and fixed browser-test generation. Learners choose the test mix and expected results, review assertions, decide any test removal, and judge the evidence. AI must not change product code or agreed behavior.

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